Anchors as Live Calibrators: Label-Free Online Conformal Uncertainty for RSSI-Based Target Tracking in Wireless Sensor Networks

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***Abstract*—Target tracking in wireless sensor networks (WSNs) based on received signal strength (RSS) reports a position estimate and, often, an uncertainty region derived from the filter covariance. Under non-line-of-sight (NLOS) propagation, environmental drift, or faulty nodes, this region becomes overconfident precisely when it matters most. Conformal prediction provides distribution-free coverage, but existing localization work calibrates on ground-truth positions collected offline, or assumes that feedback on the true target position arrives at run time. We observe that a WSN already contains nodes whose positions are known exactly: the anchors. At every time step, each anchor is localized from the remaining anchors (leave-one-anchor-out), turning the anchors into a free, live source of localization errors measured in the current radio environment. These errors, normalized by a measurement-consistency score that is available for anchors and target alike, size the conformal uncertainty region around the target, and the same inter-anchor measurements expose faulty anchors. In 600 simulation runs extending a public WSN tracking codebase, when walls appear or propagation drifts, the covariance ellipse and a one-time survey calibration contain the target only 2–15% of the time at a 90% target, whereas the proposed method maintains at least 97% without any target ground truth, at the price of regions about 25% larger than an oracle that knows the truth. A faulty anchor is detected within 4 s in all runs, with no false alarms.**

***Index Terms*—wireless sensor networks, target tracking, RSSI localization, uncertainty quantification, conformal prediction, anchor nodes, localization integrity**

I. Introduction

Localizing and tracking a moving target is a core service of wireless sensor networks (WSNs), with applications in asset tracking, worker safety, and patient monitoring [1]. Received signal strength (RSS) is the most widely available measurement, since every radio reports it at no extra hardware cost. RSS is, however, notoriously unreliable indoors: walls, furniture, and people introduce shadowing and NLOS attenuation, and the error of an RSS-based estimate varies strongly across space and time [2], [3].

For safety-related applications, the point estimate alone is not enough. A system must also say *how far off it might be*. This property, often called localization *integrity*, describes the consistency between the reported uncertainty and the actual error, and has been identified as more important than raw accuracy for location-enabled IoT, yet hard to guarantee indoors [4]. Standard trackers based on the Kalman filter [5] derive uncertainty from a covariance matrix that is only correct when the noise model is correct. When conditions change (a door closes, a crowd arrives, an anchor degrades), the covariance does not know, and the system becomes *confidently wrong*.

Conformal prediction (CP) [6], [7] offers an attractive alternative: it wraps any estimator and returns a region that contains the truth with a user-chosen probability, without assuming Gaussian noise. CP has recently been applied to indoor localization [8]–[10], and online variants adapt to distribution shift over time [11], [12]. All of these methods share one requirement: they need examples where the *true position is known*. Offline methods obtain them from costly site surveys that go stale when the environment changes [10]; online methods assume feedback on the true target position arrives during operation [12], which is not the case in a deployed tracker, since the hidden target state is precisely what is unknown [13].

Key idea. A WSN already contains devices whose positions are known exactly: the anchors. If anchors can hear one another, then at every time step each anchor can be localized from the others, exactly as the target is, and its estimation error can be measured against its known position. These *leave-one-anchor-out* residuals form a fresh calibration set, collected live, in the current radio environment, at zero labeling cost. The idea of reference nodes with known positions is old: LANDMARC [14] and related work [15] use reference tags to *correct* position estimates. We instead use them to *certify* the estimate, i.e., to drive a conformal procedure, updated at every time step, that sizes the target's uncertainty region.

Contributions.

1) A label-free, online calibration scheme for WSN tracking that uses leave-one-anchor-out residuals as the calibration set of a conformal predictor, requiring no site survey and no ground truth for the target.

2) A normalized nonconformity score that scales each error by a measurement-consistency clue computable for anchors and target alike; an ablation shows that a purely geometric normalization, although tighter in static conditions, fails under NLOS.
3) A faulty-anchor detector that exploits the known inter-anchor geometry to separate a single misbehaving receiver from environment-wide changes.
4) An open evaluation that extends a public WSN tracking codebase [16] with a realistic channel, NLOS walls, drift, and faults, and compares against covariance ellipses, survey-based conformal prediction, and oracle online conformal prediction with full and intermittent ground-truth feedback.

The remainder of the paper is organized as follows. Section II reviews related work. Section III defines the system model. Section IV presents the proposed method. Section V describes the experimental setup and results, Section VI discusses limitations, and Section VII concludes.

II. Related Work

A. RSS Localization and Tracking in WSNs

RSS-based localization maps signal strengths from anchors to a position, either through a propagation model and multilateration [1] or through learned fingerprints. Tracking chains successive estimates with a motion model, most commonly via Kalman-type filters [5]. The open-source project used as our baseline [16] learns the RSS-to-position mapping with support vector regression and smooths it with Kalman filters of several motion orders; like most such systems, it outputs only a point estimate.

B. Robustness to NLOS and Faults

A large body of work identifies and mitigates NLOS measurements [2], tolerates Byzantine anchors under NLOS bias [3], or uses trust mechanisms to limit the influence of compromised nodes during tracking [17]. These methods aim to make the *point estimate* more accurate. They do not provide calibrated statements about the residual error, which is the focus of this work.

C. Reference Nodes with Known Positions

LANDMARC [14] introduced reference tags at known positions to absorb environmental variation, and later work uses the observed offsets of reference tags to correct nearby target estimates [15]. Our work borrows the same physical insight but uses reference residuals for a different purpose: as calibration data for coverage-calibrated uncertainty regions, updated online.

D. Conformal Prediction for Localization

CP has been applied to WiFi RSS fingerprinting [8], to classification and regression variants of indoor positioning [9], and with semi-supervised calibration to reduce labeling cost [10]. These works are offline, static (no tracking), and rely on labeled calibration data. In the wireless domain more broadly, conformal calibration is advocated as a lifecycle tool for reliable AI [18].

E. Online and Weighted Conformal Prediction

Adaptive conformal inference (ACI) [11] adjusts the miscoverage level online and achieves long-run coverage under arbitrary shift. Extensions handle intermittent feedback and prior knowledge, and have been demonstrated on RSS-based localization with the feedback probability set as an experimental parameter [12]; the source of that feedback in a deployed system is left open. Weighted CP handles covariate shift [19], and online localized weighting adapts region size to test-point difficulty [20]. For hidden-state tracking, Su et al. [13] calibrate against the filter's own particle cloud because the true state is unobservable, so their guarantee concerns the filter's posterior rather than the true target position.

Gap. To the best of our knowledge, no prior work uses the anchors of a WSN as a live, label-free source of calibration data for coverage-calibrated uncertainty regions around a tracked target.

III. System Model

A. Network and Measurements

Consider K anchors at known positions $\mathbf{p}_k \in \mathbb{R}^2$, $k = 1, \ldots, K$, and a target at unknown position $\mathbf{x}_t$ at discrete time t. The RSS (dBm) received at node j from node i follows the log-distance model [21]

$$P_{ij,t} = P_0 - 10\, n \log_{10}\!\left(\frac{d_{ij,t}}{d_0}\right) + X_{ij,t} - b_{ij,t}, \qquad (1)$$

where $d_{ij,t}$ is the distance, P_0 the power at the reference distance $d_0 = 1\,\mathrm{m}$, n the path-loss exponent, $X_{ij,t} \sim \mathcal{N}(0, \sigma^2)$ log-normal shadowing, and $b_{ij,t} \geq 0$ an additional NLOS attenuation that is non-zero when the link is obstructed. A faulty anchor k reports measurements corrupted by a persistent bias or stuck value on all of its links.

We assume that anchors periodically broadcast beacons and can measure RSS from one another, so that the anchor-to-anchor RSS $\{P_{jk,t}\}_{j \neq k}$ is available at each t in addition to the target-to-anchor RSS $\{P_{k,t}\}$.

B. Localizer and Tracker

Let $f(\cdot)$ denote a snapshot localizer that maps the RSS readings from a set of anchors to a position estimate. We use weighted nonlinear least squares on (1) with the *nominal* parameters (P_0, n), solved by Gauss–Newton iterations from a weighted-centroid start; the localizer is unaware of walls, drift, and faults. Besides the estimate, f returns a context vector $\boldsymbol{\phi} = (\phi^{\mathrm{fit}}, \phi^{\mathrm{geo}})$ computable without ground truth: ϕ^{fit} is the weighted root-mean-square mismatch between the estimated and the RSS-implied distances, i.e., how strongly the readings disagree with each other, and ϕ^{geo} is the square root of the trace of the Cramér–Rao-type covariance implied by the anchor geometry. A constant-velocity Kalman filter [5] fuses the snapshot estimates $\hat{\mathbf{z}}_t$ into the tracked estimate $\hat{\mathbf{x}}_t$.

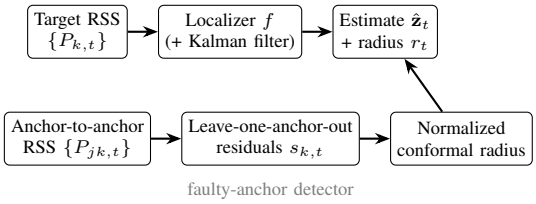


Fig. 1. Proposed pipeline. The upper path is the standard tracker. The lower path uses the anchors themselves as live calibrators to size the uncertainty region.

C. Problem Statement

Given a target miscoverage level α (here 0.1), produce at each t a region

$$\mathcal{C}_t = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x} - \hat{\mathbf{z}}_t\| \le r_t\} \tag{2}$$

such that the long-run fraction of times the true target lies outside the region stays close to α,

$$\frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\mathbf{x}_t \notin \mathcal{C}_t\} \approx \alpha, \tag{3}$$

while keeping r_t small, *without* access to the true target position at any time.

IV. Proposed Method

Fig. 1 summarizes the approach.

A. Leave-One-Anchor-Out Residuals

At each time t and for each active anchor k, the anchor is localized from the remaining anchors with the same localizer used for the target,

$$(\hat{\mathbf{p}}_{k,t},\, \boldsymbol{\phi}_{k,t}) = f\big(\{P_{jk,t}\}_{j\neq k}\big), \qquad s_{k,t} = \|\hat{\mathbf{p}}_{k,t} - \mathbf{p}_k\|. \tag{4}$$

Each $s_{k,t}$ is a genuine localization error, observed under the current propagation conditions, at no labeling cost. A sliding window of the last W steps yields up to KW calibration scores (240 for $K = 16$, $W = 15$). All K localizations share the same form and are solved jointly in one vectorized step.

B. Normalized Conformal Radius

Anchors are fixed near the walls and see different geometry and link conditions than the target, so their raw errors are not exchangeable with the target's error. We therefore use a normalized nonconformity score [7]: each residual is divided by its measurement-consistency clue,

$$\tilde{s}_{k,\tau} = \frac{s_{k,\tau}}{\phi^{\mathrm{fit}}_{k,\tau} + c}, \qquad r_t = Q_{1-\alpha}\big(\{\tilde{s}_{k,\tau}\}\big)\,\big(\phi^{\mathrm{fit}}_t + c\big), \tag{5}$$

where $Q_{1-\alpha}$ is the conformal quantile with the finite-sample correction $\lceil (n+1)(1-\alpha)\rceil/n$, τ ranges over the window, ϕ^{fit}_t is the target's own clue, and $c = 1\,\mathrm{m}$ stabilizes the ratio. When walls or drift corrupt the measurements, both the anchors' residuals and the target's clue grow, so the region widens immediately. We also evaluate the plain score ($r_t = Q_{1-\alpha}(\{s_{k,\tau}\})$), a geometric normalization by ϕ^{geo}, and an online variant in which the working level is updated as in ACI [11] using the anchors' own miscoverage as proxy feedback.

C. Faulty-Anchor Detector

Because all anchor positions are known, the nominal model predicts the RSS that anchor j should receive from anchor k. Let ρ_j be the median deviation between measured and predicted RSS over all links received by anchor j. Walls and drift shift many anchors' ρ_j together, whereas a faulty receiver deviates from the others. We compute a robust z-score of ρ_j across anchors (median and median absolute deviation) and flag anchor j when $|z_j| > 3.5$ for 5 consecutive steps; flagged anchors are excluded from both localization and calibration.

V. Experimental Evaluation

A. Baseline System and Motivation

We start from the public codebase of [16] (16 anchors on a 4×4 grid in a $200 \times 200\,\mathrm{m}$ area, SVR localization, Kalman tracking). After updating it to current Python libraries without changing its computations, it reproduces mean absolute errors of 9.35 m (SVR only), 7.14 m (SVR + Kalman), and 5.37 m (random-Fourier-feature SVR + Kalman) on its third trajectory. Inspection of the code shows that (i) the Kalman filter computes an error covariance but returns only the point estimate; (ii) the innovation covariance is formed as a product $\mathbf{CPC}^\top\mathbf{R}$ instead of the sum $\mathbf{CPC}^\top + \mathbf{R}$; (iii) the measurement-noise level is estimated once, offline, from 100 points with known positions; and (iv) the simulated channel has an effective path-loss exponent of 40, far from indoor values of 2–4 [21].

To isolate the uncertainty problem from these defects, we ran a correctly implemented constant-velocity Kalman filter on the repository's own SVR estimates, with the noise covariance measured honestly on the same 100 known points. It tracks more accurately than the original (10.66 m vs. 11.59 m mean Euclidean error), yet its nominal 90% region contains the target only 45.5% of the time (Fig. 2): the claimed radius stays near 9 m while the actual error repeatedly exceeds 30 m near the edges of the area.

B. Simulation Setup

We replace the baseline channel with (1) on a $50 \times 50\,\mathrm{m}$ indoor floor with $K = 16$ anchors on a 4×4 grid placed 3 m from the outer walls. The target follows the baseline's third trajectory scaled to the floor ($T = 660$ steps of 1 s). Nominal parameters are $P_0 = -40\,\mathrm{dBm}$, $n = 2.7$, and $\sigma = 4\,\mathrm{dB}$; each wall crossed by a link adds 8 dB of attenuation. Scenarios:

- **S1 Stationary:** nominal conditions throughout.
- **S2 Sudden NLOS:** at $t = T/2$, two walls appear and split the floor into four rooms.
- **S3 Drift:** n increases linearly from 2.7 to 3.7 over the run.
- **S4 Faulty anchor:** at $t = T/2$, one anchor's receiver acquires a $+12\,\mathrm{dB}$ bias.
- **S5 Density:** S1 and S2 with $K = 9$ and $K = 25$ anchors.

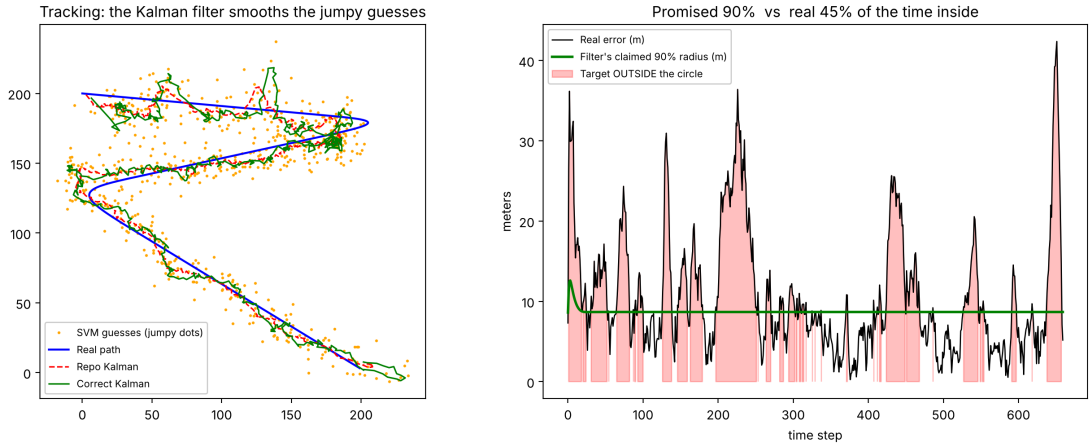


Fig. 2. Motivation on the baseline repository [16]. Left: SVR estimates (dots), true path (blue), original and corrected Kalman tracks. Right: actual error of the corrected filter (black) against its claimed 90% radius (green); shaded intervals mark the target outside the region. Empirical coverage is 45.5% instead of 90%.

Each configuration is repeated with 50 random seeds (400 runs in total), and S1–S4 are additionally repeated on a second trajectory (200 runs). We use $\alpha = 0.1$ and $W = 15$.

C. Compared Methods

All conformal methods use the same snapshot estimate $\hat{\mathbf{z}}_t$, so they differ only in how r_t is chosen.

- **Kalman ellipse (B1):** 90% ellipse of the (correct) Kalman filter around $\hat{\mathbf{x}}_t$, with measurement covariance estimated from a one-time survey.
- **One-time survey (B2):** split conformal prediction calibrated once on 100 surveyed points before tracking starts, as in [8].
- **Oracle (B3):** ACI [11] fed with the *true* target error after every step (window of 100 steps, step size 0.02). Not deployable; serves as a reference.
- **Oracle, 10% feedback (B4):** as B3, but the true error arrives only with probability 0.1, with importance-weighted updates as in intermittent online conformal prediction [12].
- **Ours:** anchor-calibrated normalized conformal radius (Section IV-B).

D. Metrics

Empirical coverage (3) over the whole run (S1) or after the change (second half, S2–S4); mean radius; recovery time, defined as the number of steps after the change (after the midpoint for the gradual drift of S3) until the 30-step rolling coverage reaches 88%; and, for S4, detection delay and false alarms. Point accuracy is unaffected by the choice of radius: the mean snapshot error is 3.3 m in S1 and rises to 13.6 m after the walls appear in S2.

E. Results

Coverage under change. Table I shows that methods calibrated once collapse when the environment changes: after the walls appear, the Kalman ellipse contains the target 4.4%

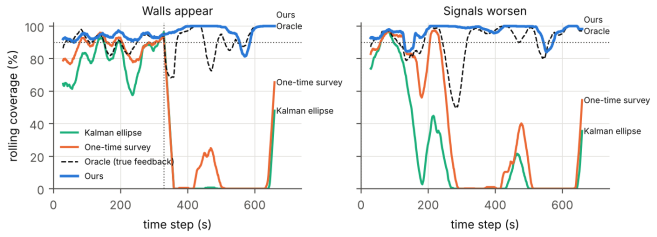


Fig. 3. Rolling coverage (30-step window, mean of 20 runs) in S2 (walls appear at the dotted line) and S3 (gradual drift). Dotted horizontal line: 90% target.

of the time and the survey radius 9.4%, and neither recovers in any of the 50 runs. The proposed method keeps 97.5% (S2) and 97.0% (S3) coverage using no target ground truth, and recovers immediately (median recovery time 0 steps in S2 and S3), whereas the oracle with 10% feedback needs a median of 31 and 46 steps and remains below target (85–86%). Fig. 3 shows the rolling coverage over time.

Price of being label-free. The method is conservative: in stationary conditions it covers 93.7% instead of 90%, with a mean radius of 6.6 m versus 5.7 m for the survey, and after a change its regions are about 25% larger than the oracle's (25.4 m vs. 20.1 m in S2). This reflects the residual mismatch between anchor and target conditions.

Ablation. Table II shows that normalization by measurement consistency reduces the over-coverage of the plain score in stationary conditions (97.0% $\rightarrow$ 93.7%) while preserving robustness. Geometric normalization is the tightest in stationary conditions (91.5%) but fails under NLOS (60.4%) because geometry cannot reveal that walls have appeared. The anchor-driven ACI correction changes results by less than one percentage point: since its feedback comes from the same anchors, it cannot correct the anchor–target mismatch.

Anchor density. With 16 or 25 anchors the method stays

TABLE I
EMPIRICAL COVERAGE (%, MEAN ± STD OVER 50 RUNS) / MEAN RADIUS (M) AT A 90% TARGET. S1: WHOLE RUN; S2–S4: AFTER THE CHANGE.

Method	S1 Stationary	S2 Walls appear	S3 Drift	S4 Faulty anchor
Kalman ellipse (B1)	84.5 ± 3.4 / 3.4	4.4 ± 0.6 / 3.4	6.0 ± 1.4 / 3.4	88.0 ± 3.8 / 3.4
One-time survey (B2)	89.0 ± 3.3 / 5.7	9.4 ± 1.4 / 5.7	10.7 ± 1.8 / 5.7	89.6 ± 3.5 / 5.7
Oracle, 10% feedback (B4)	90.4 ± 4.0 / 6.6	85.3 ± 6.3 / 19.8	86.4 ± 5.1 / 19.6	91.1 ± 4.4 / 6.8
Oracle (B3, not deployable)	89.9 ± 0.2 / 5.9	91.1 ± 0.6 / 20.1	95.3 ± 1.0 / 20.5	89.5 ± 0.6 / 5.9
Ours	93.7 ± 0.8 / 6.6	$\mathbf{97.5 \pm 0.7}$ / 25.4	$\mathbf{97.0 \pm 1.0}$ / 25.6	94.3 ± 1.3 / 6.8

TABLE II
ABLATION OF THE NONCONFORMITY SCORE (COVERAGE % / RADIUS M).

Variant	S1	S2	S3	S4
Plain	97.0/7.4	98.8/24.8	99.1/23.9	97.9/7.9
Normalized (ours)	93.7/6.6	97.5/25.4	97.0/25.6	94.3/6.8
Geometric	91.5/6.2	60.4/16.6	76.0/18.5	93.4/6.5
Plain + ACI	96.8/7.3	99.0/24.8	99.1/23.9	97.6/7.8
Normalized + ACI	93.4/6.5	97.4/25.3	96.3/25.2	93.8/6.7

TABLE III
EFFECT OF ANCHOR DENSITY (COVERAGE % / RADIUS M).

K	Method	S1 Stationary	S2 Walls appear
9	Survey	90.5/8.5	40.0/8.5
9	Oracle	90.1/8.6	91.5/16.8
9	Ours	95.4/11.0	71.3/13.4
16	Survey	89.0/5.7	9.4/5.7
16	Oracle	89.9/5.9	91.1/20.1
16	Ours	93.7/6.6	97.5/25.4
25	Survey	86.9/4.5	6.7/4.5
25	Oracle	89.7/4.8	90.9/14.2
25	Ours	92.3/5.1	94.0/16.2

above target in both conditions (Table III). With only 9 anchors it is more conservative in stationary conditions (95.4%) and falls to 71.3% after the walls appear: each anchor is localized from only 8 others and, likely, with a 3×3 grid the anchor-to-anchor links cross the new walls in a different pattern than the target's links, so the anchors under-represent the target's degradation. Sparse deployments are therefore a limitation of the approach.

Second trajectory. To check that the results do not depend on one path, we repeated S1–S4 (50 runs each) on the baseline's second trajectory, a 200-step curved path. After the change, the Kalman ellipse and the survey radius again collapse (4.1% and 9.5% in S2; 2.1% and 15.2% in S3), the oracle reaches 86–87% within the shorter remaining horizon, and the proposed method keeps 99.9% and 99.7% coverage, with regions of 27.0 m and 28.8 m versus 16.6 m and 16.1 m for the oracle; in stationary conditions it covers 92.9% (oracle 90.3%). The method is thus robust across paths but, on this path, more conservative.

Faulty anchors. In S4, the faulty anchor was detected in all 50 runs on both trajectories with a delay of 4 steps after the fault appeared (the minimum allowed by the 5-step persistence rule). No anchor was ever flagged without a fault: there were no false alarms in any run of S1–S3 (including walls and drift) or in the pre-fault halves of S4 on either trajectory, and no healthy anchor was flagged in S4.

VI. DISCUSSION AND LIMITATIONS

The central assumption is that anchor residuals represent the target's error. They do so only approximately: anchors are fixed near the walls and their links follow different paths. Our coverage results are therefore empirical rather than guaranteed, and the method compensates by being conservative (about 4 percentage points above target in stationary conditions and 25% larger regions than an oracle after a change). The approach weakens with sparse anchors (Table III). It requires anchors to exchange beacons, which costs energy and air-time in battery-powered deployments. Finally, the evaluation is simulation-based, with an idealized wall model and two trajectories; validation on real hardware and in measured RSS datasets is left for future work. When the environment degrades strongly, the radius also saturates because estimates are constrained to the floor.

VII. CONCLUSION

We proposed using the anchors of a wireless sensor network as live, label-free calibrators for the uncertainty of RSS-based target tracking. By localizing each anchor from the others at every step, the network continuously measures its own error in the current environment, and a normalized conformal procedure turns those measurements into uncertainty regions around the target. In 600 simulation runs on two trajectories, the method kept at least 97% coverage after sudden NLOS or gradual drift, where covariance ellipses and survey-based calibration fell to 2–15%, and detected faulty anchors within 4 s without false alarms. Future work includes reducing the conservativeness through better anchor–target matching, real-hardware validation, multi-target tracking, and energy-aware scheduling of anchor beacons.

CODE AVAILABILITY

The simulation code, derived from [16] and released under GPL-3.0, is available at https://github.com/lindasharaf/anchor-calibrated-wsn-tracking.

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